BIO 005 Human Physiology · Week 10 notes · Dr. Sharilyn Rennie

The heart as a pump

Need-to-know heart anatomy, smooth muscle contraction and tone, cardiac muscle and the three muscle types compared, the adrenal medulla as a sympathetic ganglion, the SA node pacemaker and the ventricular action potential, the conduction system and reading a lead II ECG, the cardiac cycle with the pressure-volume loop and heart sounds, and stroke volume, cardiac output and their control, and what happens when the pump fails. Week 10 competencies 1 to 21.

To print: these notes as a two-column PDF.

How to use these notes. Watch the lecture and work through the click-through slides first. Then read these notes to fill in anything you missed, and use them to review. The section letters match the lecture.

Your textbook: Silverthorn Chapter 15, Cardiovascular Physiology. **Using OpenStax instead?** Anatomy and Physiology 2e, Chapter 19, The Cardiovascular System: The Heart.

Need-to-know anatomy

In your textbook: Silverthorn Chapter 15, the part on the structure of the heart.

Where the heart sits

- The heart sits in the mediastinum (the middle of the chest) on the diaphragm, about two thirds of it left of the midline.
- The apex, the tip of the left ventricle, points down, forward and to the left. It lies near the fifth intercostal space at the left midclavicular line, which is where you feel the apical pulse.
- The base is the broad upper back part, formed mostly by the left atrium, where the great vessels attach.

The pericardium and the heart wall

Fibrous pericardium

A tough outer sac of dense connective tissue. It anchors the heart and keeps it from overstretching.

Serous pericardium

A thin double layer: the parietal layer lines the sac and the visceral layer covers the heart. A thin film of serous fluid between them cuts friction as the heart moves.

Epicardium

The outer layer of the heart wall (the visceral layer of the serous pericardium). The coronary vessels run in it.

Myocardium

The middle layer of cardiac muscle that does the pumping. It is most of the wall.

Endocardium

A smooth lining of endothelium over the chambers and valves, continuous with the lining of the blood vessels.

Chambers and valves

Chamber	Receives blood from	Sends blood through
Right atrium	Superior and inferior venae cavae, coronary sinus	Tricuspid valve
Right ventricle	Right atrium	Pulmonary valve to the pulmonary trunk
Left atrium	Four pulmonary veins	Mitral (bicuspid) valve
Left ventricle	Left atrium	Aortic valve to the aorta

- The atria work against low pressure and have thin walls. The right ventricle pumps to the low-pressure lungs. The left ventricle pumps against the highest pressure, so its wall is the thickest, about 10 to 15 mm.
- The two sides are two pumps in series: the output of one becomes the input of the other, so, averaged over a few beats, the two ventricles must eject the same volume.
- The AV valves (tricuspid and mitral) sit between atria and ventricles. Chordae tendineae, tendon-like cords, tie their cusps to papillary muscles in the ventricle wall.

The papillary muscles contract with the ventricle and keep the cusps from flipping back into the atrium.

- The semilunar valves (pulmonary and aortic) sit at the exits of the ventricles. Each has three cup-shaped cusps that fill with backflowing blood and snap shut.
- All four valves open and close passively, pushed by pressure differences. No nerve or muscle opens them.

Fibrous skeleton

Four rings of dense connective tissue around the valves. It anchors the valves and the muscle, and it is an electrical insulator between the atria and the ventricles.

Interventricular septum

The muscular wall between the two ventricles. The bundle of His and the bundle branches run in it.

The path of blood through the heart, step by step

1. Blood low in O2 returns from the body in the superior and inferior venae cavae to the right atrium.
2. It passes through the tricuspid valve into the right ventricle.
3. The right ventricle pumps it through the pulmonary valve into the pulmonary trunk and pulmonary arteries.
4. In the lung capillaries the blood gives up CO2 and picks up O2.
5. The pulmonary veins carry it to the left atrium.
6. It passes through the mitral valve into the left ventricle.
7. The left ventricle pumps it through the aortic valve into the aorta and out to the body.

Coronary circulation

- The right and left coronary arteries branch from the aorta just above the aortic valve. The left divides into the left anterior descending (anterior interventricular) artery and the circumflex artery.
- Coronary veins drain into the coronary sinus, which empties into the right atrium.
- Contracting muscle squeezes the coronary vessels, so the left ventricle receives most of its blood flow during diastole. A very fast heart rate shortens diastole and so shortens the time for coronary flow.
- Connections between coronary branches (anastomoses) can provide detours when one artery is partly blocked.

A. Smooth muscle: how it contracts and what controls its tone

Competencies 1 and 2: Smooth muscle contraction mechanism; Smooth muscle types and regulation.

In your textbook: Silverthorn Chapter 13, the part on smooth muscle.

Skeletal muscle needs a motor neuron and a troponin switch on the thin filament. Smooth muscle has no troponin and no sarcomeres. Its switch is on myosin and runs through an enzyme, which is why smooth muscle is slow, graded and cheap to run.

Two ways calcium rises

- Voltage route (electromechanical coupling): something depolarizes the membrane, voltage-gated L-type Ca2+ channels open, and Ca2+ enters from the extracellular fluid (ECF). Stretch, a pacemaker slow wave arriving through gap junctions, high extracellular K+, or a transmitter that opens cation channels can each do this.
- Receptor route (pharmacomechanical coupling): a hormone or transmitter binds a G protein-coupled receptor, and Ca2+ rises with no action potential at all.
- Most tissues have both routes; which one a signal uses depends on its receptors and channels. Stretch and pacemakers use the voltage route, hormones the receptor route, and nerves often both (norepinephrine also closes K+ channels, which depolarizes the cell).
- The sarcoplasmic reticulum (SR), the internal Ca2+ store, is small and releases Ca2+ through IP3 and ryanodine receptors, so smooth muscle depends heavily on Ca2+ from the ECF.

The receptor route, step by step (norepinephrine on an alpha-1 receptor)

1. Norepinephrine binds an alpha-1 receptor on the smooth muscle membrane.
2. The receptor activates its G protein, Gq.
3. Gq activates the enzyme phospholipase C.
4. Phospholipase C splits a membrane lipid and releases IP3.
5. IP3 binds channels on the SR, and stored Ca2+ flows into the cytoplasm.
6. At the same time, receptor-operated channels in the cell membrane let more Ca2+ in from the ECF.

Acetylcholine on M3 receptors, angiotensin II, oxytocin and endothelin all use this same chain, each through its

own receptor. Only the first step changes, so you learn the chain once.

Contraction and relaxation, step by step

1. Ca2+ rises in the cytoplasm through the voltage route, the receptor route, or both.
2. Four Ca2+ ions bind one calmodulin molecule.
3. The Ca2+-calmodulin complex binds and switches on myosin light chain kinase (MLCK), an enzyme that was sitting idle.
4. MLCK moves a phosphate from ATP onto the regulatory light chain of myosin. This turns on myosin ATPase.
5. Phosphorylated myosin binds actin and the crossbridges cycle. The actin is anchored to dense bodies, so the whole cell draws in like a drawstring.
6. Ca2+ is pumped out of the cell and back into the SR, and calmodulin lets go of MLCK.
7. Myosin light chain phosphatase removes the phosphate from myosin, crossbridge cycling stops, and the muscle relaxes.

Calmodulin

The Ca2+-binding protein of smooth muscle. It takes the place of troponin, but it works through myosin, not actin.

Myosin light chain phosphatase

The enzyme that takes the phosphate back off myosin. It is always active, so tone is the balance between MLCK and the phosphatase.

Dense bodies

Protein anchors in the cytoplasm and on the membrane where actin attaches. They are the smooth muscle version of Z disks.

Latch state

If the phosphate comes off while a myosin head is still bound to actin, the head can stay attached and keep holding. Tension is held with almost no ATP, which is how sphincters and artery walls hold tone for hours.

Calcium sensitization

Some signals block the phosphatase through an enzyme called Rho kinase. More myosin stays switched on at the same Ca2+ level, so tension rises without a rise in Ca2+.

Why smooth muscle is slower and cheaper to run

- Regulation is on myosin and needs an enzyme step, which takes time.
- Smooth muscle myosin ATPase is slow, so crossbridges cycle roughly 10 to 100 times more slowly than in skeletal muscle.
- The latch state holds tension for almost no ATP.
- Most of the Ca2+ comes from the ECF, so the force is graded by how much Ca2+ enters.
- With no sarcomeres to limit it, a smooth muscle cell can shorten much farther than a skeletal muscle fiber.

What changes smooth muscle tone

Signal	Example	Effect
Stretch	Higher pressure stretching an arteriole (myogenic response)	Stretch-sensitive channels open, the cell depolarizes and contracts, so flow stays steady
Local chemicals	Low O2, high CO2, H+, adenosine, nitric oxide	Relax systemic arterioles, so an active tissue gets more blood
Local chemicals	Endothelin	Contracts arterioles
Local chemicals	Histamine	Relaxes arterioles but contracts airway and gut smooth muscle
Autonomic neurons	Norepinephrine on alpha-1	Contracts vascular smooth muscle
Hormones	Epinephrine on beta-2	Relaxes vascular and airway smooth muscle
Hormones	Angiotensin II, vasopressin, oxytocin	Contract their targets (oxytocin acts on the uterus)

- Relaxing signals take a different road. Epinephrine on beta-2 raises cAMP, which turns MLCK down. Nitric oxide raises cGMP, which lowers Ca2+ and turns the phosphatase up.
- The receptor and the tissue decide the direction, not the molecule. In anaphylaxis, histamine widens arterioles (H1 receptors on the endothelium release nitric oxide, and H2 receptors on the muscle add to it), so blood pressure falls, while H1 receptors on airway muscle cause bronchospasm in the same patient.
- Calcium channel blockers close the L-type channels in vascular smooth muscle, so vessels relax and blood pressure falls.

Single-unit and multi-unit smooth muscle

	Single-unit (visceral)	Multi-unit
How cells connect	Gap junctions; the sheet contracts together	Few gap junctions; each cell has its own nerve supply
Pacemaker cells	Some cells are pacemakers	None
Control	Coarse; the tissue acts as one	Fine and graded
Examples	Gut, uterus, bladder, small blood vessels	Iris, large airways, large arteries

Many tissues have features of both types, so the two labels are a simplification. In single-unit muscle, a few cells with a nerve ending or a pacemaker can drive the whole sheet, because gap junctions pass the signal from cell to cell.

B. Cardiac muscle cells, the intercalated disc, and the three muscle types

Competencies 3 and 4: Cardiac muscle properties; Comparison of the three muscle types.

In your textbook: Silverthorn Chapter 15, the part on cardiac muscle and the heart, and Chapter 13, the comparison of the three muscle types.

Dr. Rennie's videos for this part: Contractile cells of the heart, Cardiac muscle fibers, the micro anatomy, Skeletal and cardiac muscle: how they are alike, Skeletal and cardiac muscle: how they differ, part 1, Skeletal and cardiac muscle: how they differ, part 2.

Cardiac muscle cells are short, branched and striated, and most have one central nucleus. Each cell joins its neighbors end to end at intercalated discs, and the branches let one cell touch several others. Most of what makes the heart work as a pump comes down to that junction.

Intercalated disc

The stepped junction where one cardiac cell ends and the next begins. It holds the cells together and carries the electrical signal across.

Gap junction

Channels made of connexon proteins that join the cytoplasm of two neighboring cells. Ions, and so the action potential, pass straight from one cell to the next.

Desmosome

A strong protein anchor in the intercalated disc that keeps the cells from pulling apart under the force of every beat.

Functional syncytium

A group of separate cells that behaves electrically as one unit. The atria form one and the ventricles form another. The fibrous skeleton of the heart insulates the two, so the signal crosses only through the AV node and the bundle of His.

T-tubule (cardiac)

An inward fold of the cell membrane, larger than in skeletal muscle, that carries the action potential deep into the cell.

Sarcoplasmic reticulum (cardiac)

Smaller than in skeletal muscle, so the cell holds a smaller internal Ca2+ reserve and needs Ca2+ from the ECF to trigger each beat.

Mitochondria (cardiac)

About a third of the cell's volume. They make ATP aerobically all day. At rest the heart burns mostly fatty acids, then glucose, with some lactate, and during exercise it uses more of the lactate made by skeletal muscle. Myoglobin inside the cells stores some O2. The heart cannot run up an oxygen debt, so when its blood supply is cut the cells die (myocardial infarction).

Excitation-contraction coupling, step by step

1. An action potential reaches the cell from a neighbor. Current flows through the gap junctions and depolarizes this cell to threshold. No nerve or neurotransmitter passes the signal between cardiac cells.
2. The action potential runs down the T-tubules into the cell.
3. During the plateau of the action potential, L-type Ca2+ channels (also called dihydropyridine receptors) in the T-tubule open, and a small amount of trigger Ca2+ enters from the ECF.
4. Trigger Ca2+ opens the ryanodine receptors, the Ca2+ release channels on the SR. The SR empties its store into the cytoplasm. This is calcium-induced calcium release, and the SR supplies about 90% of the Ca2+ for contraction.
5. Ca2+ binds troponin C, tropomyosin moves off the myosin-binding sites on actin, and the crossbridges

cycle. The desmosomes keep the cells attached while they pull.

6. To relax, the SR Ca2+-ATPase (SERCA) pumps Ca2+ back into the SR, and the Na+-Ca2+ exchanger moves Ca2+ out of the cell.

- In a single cardiac cell, the force is graded by how much Ca2+ is released. A single skeletal muscle twitch is all-or-none at a given fiber length.
- If the gap junctions are lost, the cells stay attached but no longer fire together, and the chamber stops acting as one unit.
- Cardiac muscle cannot be tetanized, because of its long refractory period (section D).

The three muscle types compared

	Skeletal	Cardiac	Smooth
Ca2+ source	SR only	ECF trigger through L-type channels, then the SR	Mostly ECF, some SR
Regulatory protein	Troponin, on the thin filament	Troponin, on the thin filament	Calmodulin and MLCK, on myosin
Control	Somatic motor neuron only; no nerve, no contraction	Myogenic (SA node); nerves and hormones adjust rate and force	Autonomic nerves, hormones, local chemicals, stretch; some pacemakers
Speed	Fastest	Intermediate	Slowest
Summation and tetanus	Can summate and tetanize	Cannot tetanize	Holds sustained tone at low energy cost
Gap junctions	None	Yes, in intercalated discs	Yes, in single-unit muscle
Cells	Long cylinders, many nuclei at the edge, striated	Short, branched, one central nucleus, striated	Spindle-shaped, one nucleus, not striated

C. From the spinal cord to every tissue: the adrenal medulla

Competency 5: Adrenal medulla.

In your textbook: Silverthorn Chapter 12, the part on the adrenal medulla.

The usual sympathetic pathway has two neurons in a row. In the adrenal medulla, the second neuron is replaced by a cell that secretes into the blood instead of onto one target, so its signal reaches the whole body and lasts much longer than a nerve signal.

The sympathetic signal through the adrenal medulla, step by step

1. A sympathetic preganglionic neuron in the thoracic spinal cord fires. Its axon runs to the adrenal medulla with no synapse on the way.
2. It releases acetylcholine (ACh) onto nicotinic receptors on chromaffin cells, the same way any preganglionic neuron does in a ganglion.
3. The chromaffin cells, which are modified postganglionic neurons with no axons, secrete catecholamines.
4. About 80% epinephrine and 20% norepinephrine enter the blood as hormones.
5. The blood carries them to every tissue with adrenergic receptors. That includes cells with few or no sympathetic endings, such as bronchiole smooth muscle, and the beta-2 receptors in skeletal muscle arterioles, which sit away from the nerve endings and respond mainly to circulating epinephrine. Liver cells respond by releasing glucose.
6. The effect lasts minutes, because enzymes in the liver and other tissues clear the catecholamines from the blood over several minutes.

The medulla is called a modified sympathetic ganglion because it receives preganglionic sympathetic axons and its chromaffin cells develop from the same embryonic tissue (neural crest) as postganglionic neurons. Its location next to the kidney is not the reason.

Nerve versus hormone

	Direct sympathetic nerve	Adrenal medulla
Signal	Norepinephrine released from varicosities	Mostly epinephrine, carried in the blood
Onset	Fast	Seconds
Reach	Only the tissue the nerve supplies	Every tissue with adrenergic receptors
How it ends	Reuptake into the varicosity; ends soon after firing stops	Slow clearance from the blood
How long it lasts	Brief	Minutes
What it is good for	Fast, local control	A whole-body fight-or-flight response

- The heart answers to both. Sympathetic nerves release norepinephrine onto beta-1 receptors in the heart, and circulating epinephrine reaches the same receptors.
- A pheochromocytoma is a tumor of the adrenal medulla that often releases catecholamines in bursts. Each episode of pounding heartbeat, high blood pressure and sweating lasts many minutes, because hormone in the blood is cleared slowly.

D. The SA node pacemaker and the ventricular action potential

Competencies 6, 7 and 9: Pacemaker potential; Contractile cell action potential; Cardiac refractory period.

In your textbook: Silverthorn Chapter 15, the parts on myocardial autorhythmic cells and myocardial contractile cells.

Dr. Rennie's videos for this part: What sets the pacemaker rate of the heart?, The pacemaker potential, The depolarization phase of the action potential, The repolarization phase of the action potential, How a typical cardiac cell is activated, in order.

The heart beats with no nerve telling it to, because about 1% of its cells are autorhythmic: they never rest. The fastest of these sit in the sinoatrial (SA) node. Their membrane potential has no flat resting line. As soon as one action potential ends, the cell starts drifting toward the next one.

The SA node pacemaker potential, step by step

1. At the lowest point of the cycle, about -60 mV, HCN channels open. Unlike most voltage-gated channels, they open when the cell is hyperpolarized.
2. They let mostly Na+ in (and a little K+ out). This is the funny current (I-f), and it starts the slow upward drift called the pacemaker potential.
3. As the cell nears threshold, T-type (transient) Ca2+ channels open briefly, and the entering Ca2+ finishes the climb.
4. At threshold, about -40 mV, L-type (long-lasting) Ca2+ channels open.
5. Ca2+ rushes in and the cell depolarizes to about 0 mV. SA node cells have no fast Na+ channels, so this upstroke is slower than in a ventricular cell.
6. The L-type channels inactivate and voltage-gated K+ channels open. K+ leaves, and the potential falls back toward -60 mV.
7. The K+ channels close, the HCN channels open again, and the next pacemaker potential begins.

What sets the heart rate

- The slope of the pacemaker potential sets the rate. A steeper slope reaches threshold sooner, so the heart beats faster.
- Sympathetic input raises cAMP, which opens more HCN channels and adds Ca2+ entry: the slope steepens and the rate rises.
- Parasympathetic ACh on M2 receptors opens K+ channels, which lowers the starting point, and cAMP falls: the slope flattens and the rate drops.
- The drug ivabradine blocks HCN channels, so the drift to threshold is slower and the heart rate falls.
- The SA node sets the rhythm because it reaches threshold first. Its signal fires the slower pacemakers before they reach threshold on their own.
- Nerves and hormones such as epinephrine change the timing and strength of the beat, but they do not create the rhythm. A transplanted heart, with all its nerves cut, still beats, and its resting rate runs higher, near the SA node's own rate, because vagal tone is gone.

Pacemaker	Rate on its own (beats per minute)
SA node	About 90 to 100; about 70 at rest because of vagal tone
AV node	40 to 60
Purkinje fibers	25 to 40

The ventricular contractile cell action potential, step by step

1. Phase 4, rest: the cell sits at a stable -90 mV, held there by K+ leak channels. It does not drift; it waits for a neighbor.
2. Phase 0, upstroke: current from a neighbor through gap junctions brings the cell to threshold. Fast voltage-gated Na+ channels open, Na+ pours in, and the cell reaches about +20 mV in 1 to 2 ms.
3. Phase 1, brief dip: the Na+ channels inactivate and a few K+ channels open briefly.
4. Phase 2, plateau: L-type Ca2+ channels let Ca2+ in while K+ permeability is low. The two currents balance near 0 mV for about 200 ms. This Ca2+ is the trigger for calcium-induced calcium release.
5. Phase 3, repolarization: the Ca2+ channels inactivate, slow K+ channels open, K+ leaves, and the cell returns to -90 mV.

	SA node cell	Ventricular contractile cell
Resting potential	None; lowest point about -60 mV	Stable at -90 mV
Upstroke carried by	Ca2+ through L-type channels	Na+ through fast channels
Plateau	None	About 200 ms, Ca2+ in
Fires on its own	Yes	No; waits for a neighbor

The refractory period and why the heart cannot be tetanized

- Fast Na+ channels open for the upstroke, then a second gate (the inactivation gate) closes. The channel cannot open again until the cell repolarizes.
- Because the plateau is long, the absolute refractory period lasts about 250 ms, almost as long as the contraction itself.
- A second action potential cannot start until the twitch is nearly over, so contractions cannot summate and tetanus cannot happen.
- In skeletal muscle the refractory period is over before tension peaks, which is why skeletal muscle can be tetanized.
- This matters for the pump. The ventricle has to relax to refill between beats, and a heart held in a sustained contraction would pump nothing.

E. The conduction system and the ECG

Competencies 8, 10, 11 and 12: Conduction system; ECG waves and intervals; ECG interpretation; ECG recording.

In your textbook: Silverthorn Chapter 15, the parts on electrical conduction in the heart and the electrocardiogram.

Dr. Rennie's videos for this part: What is an EKG?, The EKG waves and what each one means, The isoelectric line, Abnormalities an EKG can show, Why the ST segment matters, Why the QT interval matters, What is a stress test?, What is a Holter monitor?.

One SA node signal must reach every cell in the right order: the atria first, a short pause, then the ventricles from the bottom up. That order makes the pump efficient.

The path of one impulse, step by step

1. The SA node, in the right atrial wall near the opening of the superior vena cava, fires.
2. The signal spreads through the atrial cells by gap junctions and along the internodal pathways to the atrioventricular (AV) node. (Anatomists still debate whether these pathways are separate tracts or just faster routes through the atrial muscle.) The atria depolarize (the P wave) and then contract.
3. The fibrous skeleton, a layer of connective tissue between the atria and ventricles, blocks the signal everywhere except at the AV node and the bundle of His that leaves it.
4. The AV node, in the floor of the right atrium, holds the signal for about 0.1 s.
5. The bundle of His (AV bundle) carries the signal through the fibrous skeleton into the interventricular septum.
6. The left and right bundle branches carry it quickly down the septum toward the apex.
7. The Purkinje fibers spread it up through the ventricular walls, and the whole ventricle depolarizes within about 0.1 s (the QRS complex).
8. The ventricles contract from the apex toward the base, pushing blood up toward the semilunar valves.

The AV delay lets the atria finish contracting and emptying before the ventricles start. It happens because AV node cells are small, have few gap junctions, and use slow Ca2+ channels for their upstroke. Vagal input lengthens the delay and sympathetic input shortens it. Without the delay, the ventricles would contract while

the atria were still emptying into them, and filling would drop.

The ECG: lead II, one beat

An electrocardiogram (ECG) records the summed electrical activity of the heart from electrodes on the skin. It is not a single action potential. Lead II measures between the right arm (negative) and the left leg (positive), and in a normal heart the P wave, the tall R wave of the QRS and the T wave point upward (the small Q and S waves point down). On standard paper, one small box is 0.04 s and one big box is 0.20 s.

Part	What it records	Normal	What follows
P wave	Atrial depolarization	Under 0.12 s; one before every QRS	Atrial contraction
PR interval	Start of P to start of QRS: atria plus the AV node delay (the flat PR segment inside it is mostly the AV node delay)	0.12 to 0.20 s	The atria finish emptying
QRS complex	Ventricular depolarization; atrial repolarization is hidden inside it	About 0.06 to 0.10 s; under 0.12 s	Ventricular contraction
ST segment	All ventricular cells in their plateau	Flat, at baseline	Ejection continues
T wave	Ventricular repolarization	Upright in lead II	Ventricular relaxation
QT interval	Start of QRS to end of T: one ventricular action potential	About 0.36 to 0.44 s at a resting rate; shorter at faster rates	

- A QRS wider than 0.12 s means the signal is taking a slow route through the ventricles.
- An ST segment raised above or dropped below baseline points to injured or oxygen-starved heart muscle.
- A long QT interval raises the risk of a dangerous ventricular arrhythmia.
- Tall, peaked T waves can be a sign of high blood K+ (hyperkalemia). Flattened or inverted T waves can be a sign of ischemia, too little blood flow to the heart muscle.
- A large P wave can point to an enlarged atrium, tall R waves to a thickened ventricle, and a deep, wide Q wave to an old heart attack.

Where ECGs are used

- A 12-lead ECG combines limb and chest electrodes to record 12 views of the same beat. Each lead sees the heart from a different angle, so damage in one region shows up best in certain leads. A new tracing is compared with the patient's earlier ones whenever they exist.
- In an exercise stress test, the ECG is recorded during exercise. Narrowed coronary arteries can supply enough blood at rest but not during exercise, so ECG changes may appear only then.
- A Holter monitor is a portable ECG worn for about 24 hours or longer to catch rhythm problems or ischemia that come and go.

Reading a strip

- Heart rate: 1500 divided by the number of small boxes between two R waves, or 300 divided by the number of big boxes. For an irregular rhythm, count the QRS complexes in 6 seconds and multiply by 10.
- Example: 25 small boxes between R waves gives 1500 divided by 25, or 60 a minute.

Rhythm	What you see
Normal sinus rhythm	A P before every QRS and a QRS after every P, regular, 60 to 100 a minute
Sinus tachycardia	Sinus rhythm faster than 100 a minute
Sinus bradycardia	Sinus rhythm slower than 60 a minute; can be normal in trained athletes
First-degree heart block	Every P is followed by a QRS, but the PR interval is longer than 0.20 s
Complete (third-degree) heart block	P waves and QRS complexes march independently; the ventricles run on a slow escape rhythm

In complete block nothing gets through the AV node, so a pacemaker below the block drives the ventricles. When that escape pacemaker is in the Purkinje system, the rate is about 25 to 40 a minute with wide QRS complexes. An escape pacemaker near the AV junction runs faster, about 40 to 60, with narrower QRS complexes.

In lab: recording lead II, step by step

1. Clean and dry the skin. Place the negative electrode on the right arm and the positive electrode on the left leg (or lower left torso), and the ground electrode where your kit directs.

2. Have the subject sit or lie still and relaxed (muscle movement adds noise) and record several beats.
3. Measure the PR interval, QRS duration and QT interval by counting small boxes (0.04 s each) or with the software cursors.
4. Calculate the heart rate from the spacing of the R waves.
5. Match each wave to the mechanical event that follows it: P then atrial contraction, QRS then ventricular contraction and S1, T then ventricular relaxation, with S2 near the end of the T wave.

F. The cardiac cycle, the pressure-volume loop, and heart sounds

Competencies 13, 14, 15 and 21: Cardiac cycle; Pressure volume loop; Heart sounds; Heart sound and pulse correlation.

In your textbook: Silverthorn Chapter 15, the part on the heart as a pump: the cardiac cycle, pressure-volume curves and heart sounds.

Dr. Rennie's videos for this part: One cardiac cycle, with the EKG waves and the mechanical events, Systole and diastole, Cardiac cycle, step 1: ventricular filling, End-diastolic volume and what sets it, Cardiac cycle, step 2: isovolumetric contraction, Cardiac cycle, step 3: ventricular ejection, Cardiac cycle, step 4: isovolumetric relaxation, The cardiac cycle, all together, Important points about the cardiac cycle, The left heart and the right heart, End-systolic volume, What is auscultation?, The typical heart sounds, Valve closure, heart sounds and the pressure changes, Where to listen for each heart sound, The normal heart sounds and the phases they belong to, Heart murmurs.

Blood flows from higher pressure to lower pressure, and the valves open and close passively because of the pressure on each side. An AV valve is open when atrial pressure is higher than ventricular pressure. A semilunar valve opens only when ventricular pressure rises above the pressure in its artery.

Systole

The contraction phase of a chamber.

Diastole

The relaxation phase of a chamber.

End-diastolic volume (EDV)

Volume at the end of filling, about 135 mL at rest.

End-systolic volume (ESV)

Volume left after ejection, about 65 mL at rest.

Isovolumic

Same volume. All four valves are closed, so the pressure changes but the volume cannot.

The cardiac cycle, phase by phase

1. Atrial systole: the atria contract and top up the ventricles to EDV. This adds the last part of filling, about 20% at rest. AV valves open, semilunar valves closed.
2. Isovolumic ventricular contraction: the ventricles begin to contract and their pressure rises above atrial pressure, so the AV valves close (S1). Pressure is still below aortic pressure, so all valves are closed and the volume stays fixed.
3. Ventricular ejection: left ventricular pressure passes aortic pressure (about 80 mm Hg) and right ventricular pressure passes pulmonary trunk pressure. The semilunar valves open and blood leaves. Left ventricular pressure peaks near 120 mm Hg, and the volume falls to ESV.
4. Isovolumic ventricular relaxation: the ventricles relax and their pressure drops below aortic pressure, so the semilunar valves close (S2). Pressure is still above atrial pressure, so all valves are closed again.
5. Ventricular filling: ventricular pressure drops below atrial pressure, the AV valves open, and blood flows in passively. Most filling happens in this phase.

The right side goes through the same phases at the same time with lower pressures. The right ventricle peaks near 25 mm Hg, yet it ejects the same volume per beat as the left.

- Timing at about 75 beats per minute: atrial systole lasts about 0.1 s, ventricular systole about 0.3 s, and the rest of the cycle, when all chambers are relaxed, about 0.4 s.
- When the heart rate rises, mostly the relaxation time shortens. Systole shortens only a little, so at very high rates filling time is what runs out.
- When the aortic valve snaps shut, blood rebounds off the closed cusps and makes a small bump on the aortic pressure tracing, the dicrotic notch.
- Each ECG wave comes just before the mechanical event that follows it: P before atrial systole, QRS before ventricular systole, T before relaxation.

The left ventricular pressure-volume loop

The loop plots left ventricular pressure (up) against volume (across) for one beat. It runs counterclockwise, and its four corners are the four valve events. Values are typical textbook numbers at rest.

Part of the loop	Event	Typical value
Bottom, left to right	Filling; mitral valve open	Volume rises from 65 to 135 mL at low pressure
Corner A, bottom right	Mitral valve closes (S1)	EDV, about 135 mL: the preload
Right side, straight up	Isovolumic contraction	Volume fixed, pressure rising
Corner B, top right	Aortic valve opens	About 80 mm Hg: the afterload
Top, right to left	Ejection	Pressure peaks near 120 mm Hg
Corner C, top left	Aortic valve closes (S2)	ESV, about 65 mL
Left side, straight down	Isovolumic relaxation	Volume fixed, pressure falling
Corner D, bottom left	Mitral valve opens	Filling starts again

The width of the loop is the stroke volume, EDV minus ESV.

Change	What moves on the loop	Stroke volume
More preload	Corner A moves right; the loop widens	Rises (Frank-Starling)
More afterload	Corner B moves up; ejection ends sooner, so corner C sits farther right; the loop is taller and narrower	Falls
More contractility	Pressure peaks higher and corner C moves left; the loop widens	Rises
Failing ventricle	A short, narrow loop	Falls

Heart sounds

S1 (lub)

The AV valves (mitral and tricuspid) closing at the start of ventricular systole. Loudest at the apex.

S2 (dup)

The semilunar valves (aortic and pulmonary) closing at the start of ventricular diastole. Loudest at the base of the heart, in the second intercostal spaces.

Murmur

An extra sound made by turbulent blood flow. Smooth flow is silent.

Stenosis

A valve that does not open fully, so blood is forced through a narrow opening.

Regurgitation

A valve that does not close fully, so blood leaks backward. Also called insufficiency or incompetence.

- S1 is a little longer and lower in pitch than S2. S1 is loudest at the apex and S2 at the base. The sounds come from the vibration and turbulence set off as the valves snap shut.
- A third and fourth heart sound also exist: S3 during rapid early filling and S4 when the atria contract. Both are usually too quiet to hear in an adult, although an S3 can be normal in children and young adults.
- In mitral valve prolapse, one or both mitral cusps bulge back into the left atrium during systole. It is common and often harmless, but it can let blood leak back (mitral regurgitation).
- Rheumatic fever, an immune reaction after a strep throat infection, can scar the valves and cause stenosis or regurgitation years later.
- A pericardial friction rub is a scratchy sound made when inflamed pericardial layers rub together (pericarditis). It is not a valve sound.

Timing	Possible causes	Classic examples
Systolic: between S1 and S2	An AV valve leaking, or a semilunar valve too narrow	Mitral regurgitation at the apex; aortic stenosis at the right second intercostal space
Diastolic: after S2, before the next S1	A semilunar valve leaking, or an AV valve too narrow	Mitral stenosis at the apex; aortic regurgitation at the base

Valve areas on the chest wall

Sound travels in the direction the blood flows, so each valve is heard best a little downstream of where it sits.

Area	Where to listen	What you hear best
Aortic	Second intercostal space, right sternal border	S2; aortic stenosis murmur
Pulmonic	Second intercostal space, left sternal border	S2 from the pulmonary valve
Tricuspid	Fourth or fifth intercostal space, left lower sternal border	S1 from the right side; tricuspid regurgitation
Mitral (apex)	Fifth intercostal space, left midclavicular line	S1 loudest; the apical pulse; mitral murmurs

In lab: heart sounds and the pulse, step by step

1. In a quiet room, listen at each of the four areas in turn with the diaphragm of the stethoscope.
2. Pick out S1 and S2. At a resting rate, the gap from S1 to S2 (systole) is shorter than the gap from S2 to the next S1 (diastole).
3. Feel the radial pulse while you listen. It arrives just after S1, during ejection, a little late because the pressure wave takes time to reach the wrist.
4. Count the apical and radial rates over the same minute. They are normally equal; a lower radial count (a pulse deficit) means some beats are too weak to reach the wrist.
5. Match the sounds to the cycle: S1 starts isovolumic contraction, the pulse comes during ejection, S2 starts isovolumic relaxation.

G. Stroke volume, cardiac output, and what controls them

Competencies 16, 17, 18, 19 and 20: Stroke volume and ejection fraction; Cardiac output; Frank Starling relationship; Preload afterload and contractility; Autonomic regulation of the heart.

In your textbook: Silverthorn Chapter 15, the part on cardiac output and the factors that influence heart rate and stroke volume.

Dr. Rennie's videos for this part: Substances that change contraction: inotropy and chronotropy, Cardiac output and what determines it, Cardiac reserve, Three factors that set stroke volume and keep both ventricles equal, Summary tables: what changes cardiac output.

The numbers for a typical resting adult

Measure	Formula	Typical value
Stroke volume (SV)	EDV minus ESV	135 minus 65 = 70 mL per beat
Ejection fraction (EF)	SV divided by EDV, × 100	70 divided by 135 ≈ 52%
Cardiac output (CO)	Heart rate × SV	72 × 70 mL = 5,040 mL/min, about 5 L/min
Cardiac reserve	Maximum CO minus resting CO	About 15 to 20 L/min in a healthy young adult

- A normal ejection fraction is about 50 to 70% (the exact cutoffs vary a little between sources). The textbook numbers above give 52%, near the low end of normal.
- An ejection fraction of 40% or lower is a reduced ejection fraction: the ventricle is not emptying well, which is one form of heart failure. Values of 41 to 49% are called mildly reduced. Heart failure can also happen with a normal ejection fraction when a stiff ventricle fills poorly.
- A resting cardiac output of about 5 L/min moves roughly the whole blood volume through the heart each minute.
- A healthy young adult can reach a cardiac output of about 20 to 25 L/min in hard exercise, about four to five times the resting value. Top endurance athletes can reach seven to eight times their resting output. In heart failure there is little or no reserve, so ordinary daily activities become hard.
- Cardiac output changes in proportion to heart rate or stroke volume. Doubling the heart rate with the same stroke volume doubles the output. Exercise raises both at once. Anything that lowers the maximum heart rate or stroke volume (a beta blocker, a weak ventricle) shrinks cardiac reserve, and a raised resting output (as in fever) uses part of it up.
- There is a limit. At a very fast heart rate, diastole becomes so short that the ventricle cannot fill, stroke volume falls, and cardiac output can drop.

State	Stroke volume	Heart rate	Cardiac output
Rest	70 mL	75 a minute	5,250 mL/min, about 5.3 L/min
Mild exercise	100 mL	100 a minute	10,000 mL/min, 10 L/min
Intense exercise	130 mL	150 a minute	19,500 mL/min, about 19.5 L/min

These are practice numbers; real values vary with body size, sex and fitness. Exercise, fever and thyroid hormone all raise cardiac output.

Practice: EDV 150 mL, ESV 90 mL, heart rate 80 gives SV 60 mL, EF 40% (reduced) and CO 4.8 L/min.

The Frank-Starling law

The more the ventricle is stretched by filling (the larger the EDV), the harder it contracts on the next beat, so stroke volume rises to match venous return. This is built into the muscle and needs no nerve. It keeps the two sides of the heart matched beat by beat, so blood does not pile up in the lungs or in the body.

- Why it works: at rest, cardiac sarcomeres are shorter than their best length. Stretch moves actin and myosin overlap closer to the best length, more crossbridges can form, and the filaments become more sensitive to Ca2+.
- Venous return, the blood coming back to the heart, sets EDV. The skeletal muscle pump, the respiratory pump, sympathetic constriction of the veins, and a larger blood volume all raise it.
- Example: lying down suddenly raises venous return. The next EDV is larger, the fibers stretch, and stroke volume rises with no nerve input.

Preload, afterload and contractility

Preload

How much the ventricle is stretched before it contracts. EDV is used as the measure.

Afterload

The load the ventricle works against to eject blood, mainly the arterial pressure it must beat to open the semilunar valve.

Contractility (inotropy)

The force of contraction at a given EDV. It depends on how much Ca2+ reaches the contractile proteins.

Chronotropy

An effect on heart rate. A positive chronotrope speeds the heart; a negative one slows it.

Factor	What raises or lowers it	Effect on stroke volume
More preload	More venous return, longer filling time	Rises
More contractility	Sympathetic input, epinephrine, digoxin	Rises
Less contractility	Beta blockers, some calcium channel blockers, low O2, acidosis	Falls
More afterload	High blood pressure, aortic stenosis	Falls at the same force of contraction

Sympathetic input at the receptor level, step by step

1. Norepinephrine from sympathetic nerves, or epinephrine from the adrenal medulla, binds beta-1 receptors on SA node cells and contractile cells.
2. The receptor activates its G protein, Gs, which activates adenylyl cyclase. cAMP rises in the cytoplasm.
3. In SA node cells, cAMP binds the HCN channels and opens more of them, and more Ca2+ enters. The pacemaker potential steepens and the rate rises (positive chronotropy).
4. In contractile cells, cAMP activates protein kinase A (PKA). PKA phosphorylates the L-type Ca2+ channels, more Ca2+ enters, the SR releases more Ca2+, and the contraction is stronger (positive inotropy).
5. PKA also phosphorylates phospholamban, a protein that normally slows SERCA. SERCA then pumps Ca2+ back into the SR faster. PKA also phosphorylates troponin I, which lets Ca2+ come off troponin sooner. Both make the muscle relax sooner, so the ventricle still has time to fill at high heart rates.

Parasympathetic input at the receptor level, step by step

1. The vagus nerve releases ACh onto M2 muscarinic receptors in the SA node and AV node.
2. The receptor activates its G protein, Gi. Gi opens K+ channels, which hyperpolarizes the cell, and it inhibits adenylyl cyclase, so cAMP falls and fewer HCN and Ca2+ channels open.
3. Less funny current and a lower starting point flatten the pacemaker potential, so the heart rate falls. In the AV node, conduction slows and the PR interval lengthens.

4. Few vagal fibers reach the ventricular muscle, so parasympathetic input has little direct effect on ventricular force. Vagal fibers do reach the atrial muscle, where ACh weakens atrial contraction.

- A beta-1 blocker stops norepinephrine and epinephrine from raising cAMP in both cell types, so heart rate and contractility both fall.
- The two divisions are both active at rest, and the body shifts the balance between them. Strong vagal firing can slow the heart well below its resting rate.

Where the commands come from

1. Sensors report to the cardiovascular center in the medulla oblongata. Baroreceptors in the aortic arch and carotid arteries sense stretch from blood pressure. Chemoreceptors sense changes in blood chemistry such as O2, CO2 and H+. Proprioceptors in limbs and muscles sense movement, and they help drive the quick rise in heart rate when exercise starts.
2. The cardiovascular center changes how often the sympathetic and parasympathetic neurons fire.
3. Sympathetic signals travel down the spinal cord to the thoracic region and out along the cardiac accelerator nerves to the SA node, the AV node and most of the myocardium.
4. Parasympathetic signals travel in the right and left vagus nerves (cranial nerve X) to the SA node, the AV node and the atrial muscle.

Other things that change heart rate and force

Factor	Effect
Adrenal epinephrine and norepinephrine	Raise rate and contractility in exercise, stress and excitement
Thyroid hormone	Raises rate and contractility; a fast resting rate is a sign of hyperthyroidism
High blood K+	Lowers rate and force; it depolarizes the resting cells, which hampers action potentials, and can cause dangerous arrhythmias
Ca2+ in the ECF	A moderate rise strengthens the beat; Ca2+ levels outside the normal range disturb the rhythm
Low O2, acidosis or alkalosis	Depress heart activity
Age	Newborns have resting rates near 120 or more; the resting rate falls through childhood
Sex	Adult females tend to have a slightly higher resting rate
Fitness	Training lowers the resting rate; athletes may rest below 60 because each beat ejects more
Body temperature	Fever and heat raise the rate; cooling slows the rate and weakens the beat

Surgeons sometimes cool a patient on purpose during heart surgery. The low temperature slows metabolism and O2 demand, so the heart and other tissues tolerate reduced blood flow for longer.

Clinical connections: when the pump fails

In your textbook: Silverthorn Chapter 15, the parts on myocardial contractile cells, the cardiac cycle and cardiac output.

Coronary artery disease and myocardial infarction

1. LDL cholesterol collects in the wall of a coronary artery, inflammation draws in white blood cells, and an atherosclerotic plaque with a fibrous cap forms and narrows the artery.
2. Blood flow may be enough at rest but not when O2 demand rises, as in exercise. The muscle beyond the narrowing becomes ischemic (short of blood flow and O2).
3. Ischemia can cause angina pectoris, a squeezing chest pain that may spread to the neck, jaw or left arm. Some ischemic episodes cause no pain at all (silent ischemia).

4. If the plaque's cap breaks open, a clot forms on it and can block the artery completely.
5. Cardiac cells depend on aerobic ATP, so the cells beyond a complete block die. This is a myocardial infarction (heart attack).
6. Dead muscle is replaced by scar tissue that does not contract, so the ventricle weakens. Damaged tissue can also disturb conduction and trigger dangerous arrhythmias such as ventricular fibrillation.

- Injured heart cells leak their contents into the blood. Cardiac troponin is the main blood test for heart muscle damage. CK-MB, the heart form of the enzyme creatine kinase, is an older marker. Myoglobin rises early but also comes from skeletal muscle, so it is not specific to the heart.
- An infarct often shows on the ECG as a raised ST segment, and later as Q waves. ST depression points to ischemia.
- Restoring flow quickly limits damage, though returning O2 can itself harm cells through free radicals (reperfusion injury).
- Risk factors include smoking, high blood pressure, diabetes, high LDL cholesterol, obesity, inactivity and family history; many can be changed.

Heart failure

In heart failure a ventricle cannot pump out the blood it receives as fast as it arrives, so blood backs up behind it. Cardiac reserve is small, and one side usually fails first.

	Left-sided failure	Right-sided failure
Where blood backs up	Into the pulmonary veins and lungs	Into the systemic veins
What you see	Pulmonary edema: fluid in the lungs and shortness of breath	Peripheral edema: swelling in the feet, ankles and legs

- Lowering afterload helps a weak ventricle eject more blood, which is one reason blood pressure control matters in heart failure.
- The kidneys respond to the low output by holding on to salt and water. Blood volume rises, which adds to the edema over time.

Cardiac tamponade

1. Fluid or blood collects in the pericardial cavity, for example from pericarditis or an injury.
2. The fibrous pericardium cannot stretch, so the fluid squeezes the heart and the ventricles cannot fill: EDV and venous return fall.
3. Stroke volume and cardiac output fall, and blood pressure drops.
4. Draining the fluid through a needle (pericardiocentesis) lets the heart fill again.

Congenital heart defects

These are present at birth. Many are mild and some go unnoticed for life; others need surgery. Most change the pressures or let blood take a shortcut (a shunt) between the two sides.

Patent ductus arteriosus

The fetal vessel between the pulmonary trunk and the aorta stays open after birth. Higher aortic pressure pushes some blood back into the pulmonary trunk, so the lungs get extra flow and the heart works harder.

Septal defect

A hole in the wall between the atria (often a foramen ovale, the fetal opening, that did not close) or between the ventricles. Blood usually shunts from the higher-pressure left side to the right.

Coarctation of the aorta

A narrowed segment of the aorta. Afterload rises, the left ventricle works harder, and blood pressure is high in the arms and lower in the legs.

Tetralogy of Fallot

Four defects together: a ventricular septal defect, an aorta that sits over both ventricles, a narrowed pulmonary valve, and a thickened right ventricle. Blood low in O2 reaches the aorta, so the baby looks blue (cyanosis).

Inflammation of the heart wall

- Pericarditis: inflamed pericardium, often after a virus; chest pain and a friction rub.
- Myocarditis: inflamed heart muscle, usually after a virus; fatigue, chest pain, a fast or irregular beat.
- Endocarditis: inflamed endocardium, usually bacterial infection of the valves; fever with a new murmur.